Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Main Full Syllabus Test - 5

Test Date: 30th Mar 2025

INSTRUCTIONS

- The test is of 3 hours duration.
- The Test Booklet consists of 75 questions. The maximum marks are 300.
- There are three parts in the question paper (1, 2, 3) consisting of Physics, Chemistry and Mathematics, having 25 questions in each subject and each subject having two sections.
  (I) Section-A contains 20 multiple choice questions with only one correct option.
  (II) Section-B contains 5 numerical value type questions.
- Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.
- The answer should be within 0 to 9999. If the answer is in decimal, then round off to the nearest integer value (e.g., If the answer is above 10 and less than 10.5, round off to 10; if the answer is more than or equal to 10.5 but less than 11, round off to 11).
- Use blue/black ballpoint pen only for writing particulars or marking responses on the answer sheet. Use of pencil is strictly prohibited.

MATHEMATICS

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let the position vectors of the points A, B, C and D be $5\hat{i}+5\hat{j}+2\lambda\hat{k}$, $\hat{i}+2\hat{j}+3\hat{k}$, $-2\hat{i}+\lambda\hat{j}+4\hat{k}$ and $-\hat{i}+5\hat{j}+6\hat{k}$ Let the set $S=\{\lambda\in R;$ the points A, B, C and D are coplanar $\}$ then $\sum_{\lambda\in S}(\lambda+3)^2$ is equal to

1) $\frac{37}{2}$ **2)**13 **3)** 61 **4)** 41

2. Let $P(x_0,y_0)$ be the point on the hyperbola $3x^2-4y^2=36,$ which is nearest to the line $2y+3x=1$ then $\sqrt{2}(y_0+x_0)$ is equal to

1) 9 **2)** -3 **3)** -9 **4)** 3

3. Given: A circle, $2x^2+2y^2=5$ and a parabola $y^2=4\sqrt{5}x$

Statement 1 : An equation of a common tangent to these curves is y= $x+\sqrt{5}$

Statement 2: If the line $y=mx+\frac{\sqrt{5}}{m}(m\neq 0)$ is their common tangent, then m satisfies $m^4-3m^2+2=0$

1)Statement 1 is true Statement 2 is true

2)Statement 1 is true, Statement 2 is false

3)Statement 1 is false, Statement 2 is true

4)Statement 1 is false statement 2 is false

4. If $x^3-ax^2+bx+c=0$ has the roots $\alpha^2+\beta^3+\gamma^4,\beta^2+\gamma^3+\alpha^4$ and $\gamma^2+\alpha^3+\beta^4$ where α,β,γ are the roots of $x^3-x^2-1=0$ then the value of $2a+b+c$ is equal to

1)14 **2)**15 **3)**17 **4)**16

5. Let the number of $(22)^{2022}+(2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7 then $\left(\alpha^3+\beta^3\right)$ is equal to

1)5 **2)**9 **3)**17 **4)**20

6. Match the following

	Column I		Column II
A)	Let f be a real valued differentiable function on R such that $f'(1)=6$ and $f'(2)=2$ then $\lim_{h\to 0}\frac{f(3\cosh+4\sinh-2)-f(1)}{f(3e^h-5\sec h+4)-f(2)}$ is	**P)**	4
B)	Let $a>0$ let $f:[-4a,4a]\to R$ be an even function such that	**Q)**	5

	$f(x)=f(4a-x)\forall x\in[2a,4a]$ and $\lim_{h\to0}\frac{f(2a+h)-f(2a)}{h}=4$ then $\lim_{h\to0}\frac{f(h-2a)-f(-2a)}{h}$ is equal to		
C)	Suppose f is differentiable function on R Let $F(x)=f(e^x)$ and $G(x)=e^{f(x)}$. If $f'(1)=e^3$ and $f(0)=f'(0)=3$ then $\frac{G'(0)}{f'(0)}$ is equal to	**R)**	3
D)	Let $f(x)=\max\{\cos x, x, 2x-1\}$ where $x>0$ then number of points non – differentiability of f(x) is equal to	**S)**	2
		T)	1

1) $A-P, B-R, C-S, D-S$ **2)** $A-P, B-S, C-R, D-S$

3) $A-P, B-Q, C-R, D-S$ **4)** $A-P, B-S, C-T, D-R$

7. Let $y=y(x)$ be the solution curve of the differential equation

$\frac{dy}{dx}=\frac{y}{x}\left(1+xy^2\left(1+\log x\right)\right), x>0,\ y(1)=3$ then $\frac{y^2(x)}{9}$ is equal to

1) $\frac{x^2}{5-2x^3\left(2+\log_e x^3\right)}$ **2)** $\frac{x^2}{2x^3\left(2+\log_e x^3\right)-3}$

3) $\frac{x^2}{3x^3\left(1+\log_e x^3\right)-2}$ **4)** $\frac{x^2}{7-3x^3\left(2+\log_e x^2\right)}$

8. An organization awarded 48 medals in event 'A', 25 in event 'B' and 18 in event 'C'. If these medals went to total 60 men and only five men got medals in all the three events, then how many received medals in exactly two of three events?

1)10 **2)**9 **3)**21 **4)**15

9. Let $S=\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+.....+\sqrt{1+\frac{1}{1999^2}+\frac{1}{2000^2}}$ then the $\left|2000\left(S-2000\right)\right|=.....$

1)3 **2)**2 **3)**1 **4)**5

10. Let set A and B have 5 elements each. Let mean of the elements in sets A and B be 5 and 8 respectively and the variance of the elements in sets A and B be 12 and 20 respectively. A new set C of 10 elements is formed by subtracting 3 from each element of A and adding 2 to each element of B. Then the sum of the mean and variance of the elements of C is ……

1)36 **2)**38 **3)**32 **4)**40

11. Consider the equation $\frac{xy}{x+y}=2^3 3^4 5^6$ then the number of positive integral solutions of the equation are

1) 140 **2)**819 **3)**72 **4)**601

12. The volume of greatest cone obtained by rotating a right angle triangle of hypotenuse 2 about a side is $\frac{k\pi}{9\sqrt{3}}$ cubic units, then the value of k is

1) 16 **2)** 9 **3)** 7 **4)** 4

13. Let $\alpha,\beta,\gamma \in R$, Let the mean and the variance of 6 observations -3, 4, 7, -6, α,β be 2 and 23 respectively. The mean deviation about the mean of these 6 observations is……

1)$\frac{13}{3}$ **2)**$\frac{16}{5}$ **3)**$\frac{11}{3}$ **4)**$\frac{14}{3}$

14. The value of $\int_0^1 \left(2x^3-3x^2-x+1\right)^{1/3}$ dx is equal to

1)-1 **2)**2 **3)**0 **4)**1

15. Let d be the distance of the point of intersection of the lines $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ and $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ from the point (7, 8, 9) then d^2+6 is equal to

1)72 **2)**69 **3)**75 **4)**78

16. A bag contains 24 identical balls of which 12 are white and 12 black. The balls, are drawn at random from the box one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is

1)$\frac{5}{64}$ **2)**$\frac{27}{32}$ **3)**$\frac{5}{32}$ **4)**$\frac{1}{2}$

17. The area (in sq units) of the region $A=\left\{(x,y)\in R\times R / 0\le x\le 3,\ 0\le y\le 4,\ y\le x^2+3x\right\}$ is

1)$\frac{59}{6}$ **2)**8 **3)**$\frac{53}{6}$ **4)**$\frac{26}{3}$

18. The system of linear equations $\lambda x+2y+2z=5;\ 2\lambda x+3y+5z=8;\ 4x+\lambda y+6z=10$ has

1)No solution when $\lambda=8$ **2)**a unique solution when $\lambda=-8$

3)infinitely many solution when $\lambda = 2$ **4)**No solution when $\lambda = 2$

19. If $\alpha = \tan\left(\frac{5\pi}{16}\left(\sin\left(2\cos^{-1}\frac{1}{\sqrt{5}}\right)\right)\right)$ and $\beta = \cos\left[\sin^{-1}\left(\frac{4}{5}\right)+\sec^{-1}\left(\frac{5}{3}\right)\right]$ then the inverse trigonometric function take principle values then the equation whose roots are α and β is

1) $25x^2+18x+7=0$ **2)** $25x^2-18x+7=0$

3) $25x^2-18x-7=0$ **4)** $25x^2+18x-7=0$

20. If $A=\{1,2,3\}, B=\{1,2,3,4,5,6,7,8\}$ then the number of functions f from A to B such that $f(i)\le f(j)$ whenever $i<j$ is

1)150 **2)**120 **3)**180 **4)**240

SECTION-II(NUMERICAL VALUE TYPE)

This section contains **5Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** andIf answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. If A and B are square matrices of order 3 such that $2(A+B)=A^T+B^T+3I$ and $AA^T=4I$ then $\left|12A^{-1}-BA^T+I\right|=$……..

22. Let $Z=1+i$ and $Z_1=\frac{1+i\bar{Z}}{\bar{Z}(1-Z)+\frac{1}{Z}}$ then $\frac{36}{\pi}\arg(Z_1)$ is……

23. Let a ray of light passing through the point (3, 10) reflects on the line $2x+y=6$ and the reflected ray passes through the point (7,2). If the equation of the incident ray is $ax+by+1=0$ then a^2+b^2+3ab is equal to

24. Let e_1 and e_2 be the eccentricities of the ellipse , $\frac{x^2}{25}+\frac{y^2}{b^2}=1$ ($b<5$) and the hyperbola, $\frac{x^2}{16}-\frac{y^2}{b^2}=1$ respectively satisfying $e_1e_2=1$. If α and β are distances between the foci of ellipse and foci of the hyperbola respectively, then $5\alpha+4\beta$ is equal to

25. **If** $\int\frac{\sin x}{\sin^3 x+\cos^3 x}dx=\alpha.\log_e\left|1+\tan x\right|+\beta\left|1-\tan x+\tan^2 x\right|+\gamma.\tan^{-1}\left(\frac{2\tan x-1}{\sqrt{3}}\right)+C$ where C is constant of integration, then the value of $18\left(\alpha+\beta+\gamma^2\right)=$.....

PHYSICS **Max Marks: 100**

SECTION-I(SINGLE CORRECT ANSWER TYPE)

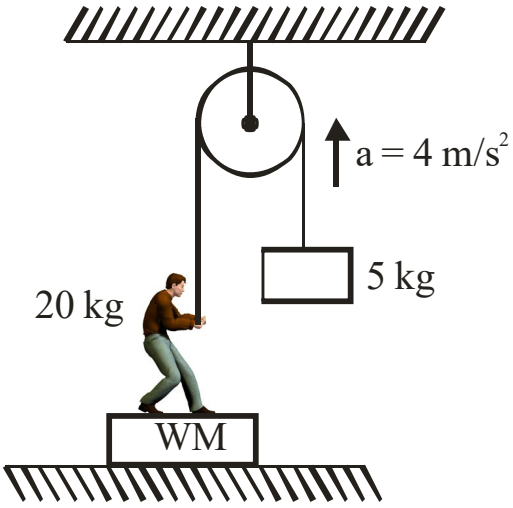
This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. Twelve persons are initially at the twelve corners of a regular polygon of twelve sides of side a. Each person now moves with uniform speed V in such a manner that 1 is always directed towards 2, 2 towards 3, 3 towards 4 and so on. The distance travelled by each person before they meet is

1) $\frac{2a}{2+\sqrt{3}}$ **2)** $\frac{2a}{2-\sqrt{3}}$ **3)** $\frac{2a}{\sqrt{3}}$ **4)** $\frac{a}{2+\sqrt{3}}$

27. Boy is pulling the rope as shown below. Reading of the weighing machine is

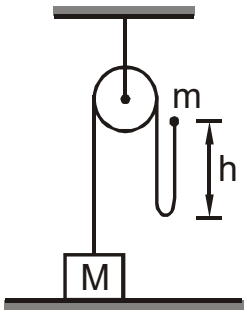


1)130 N **2)**70 N **3)**100 N **4)**280 N

28. A point Transversed 3/4th of the circle of radius R in time t. The magnitude of the average velocity of the particle in this time interval is

1) $\frac{\pi R}{t}$ **2)** $\frac{3\pi R}{2t}$ **3)** $\frac{R\sqrt{2}}{t}$ **4)** $\frac{R}{\sqrt{2}\,t}$

29. Two masses M and m are tied with a string and arranged as shown. The velocity of block M when it loses the contact is (mass 'm' will not collide with horizontal plane in course of motion)



1) $2\sqrt{gh}$ **2)** $\frac{m\sqrt{gh}}{(m+M)}$ **3)** $\frac{2m\sqrt{gh}}{(m+M)}$ **4)** $\frac{2M\sqrt{gh}}{(m+M)}$

30. **Assertion (A)**: The net momentum of system of particles in centre of mass frame is always zero.

Reason (B) : The momentum of centre of mass of system of particles is always equal sum of momentum of particles.

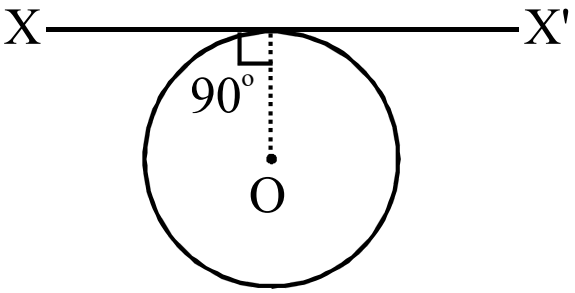
1) Assertion is True, Reason is True; Reason is **a Correct** explanation for Assertion

2) Assertion is True, Reason is True; Reason is **a not correct** explanation for Assertion

3) Assertion is True, Reason is False

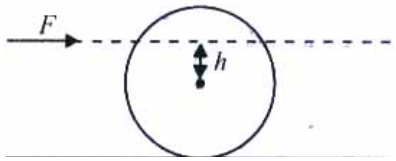
4) Assertion is False, Reason is True

31. A thin wire of length ℓ and uniform linear mass density ρ is bent into a circular loop with center at O an shown. The moment of inertia of the loop about the axis XX′ in the plane of the loop is



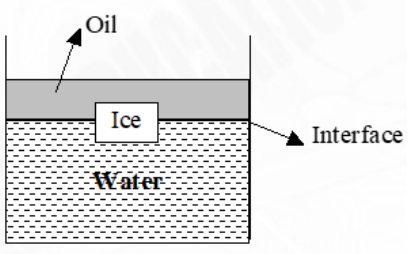
1) $\frac{\rho L^3}{8\pi^2}$ **2)** $\frac{\rho L^3}{16\pi^2}$ **3)** $\frac{5\rho L^3}{16\pi^2}$ **4)** $\frac{3\rho L^3}{8\pi^2}$

32. A billiard ball, initially at rest on a smooth horizontal surface is given a sharp impulse by a rod. The rod is held horizontally at a height h above the centre of the ball. The ball immediately begins to roll without slipping after the impact. Calculate the height h in terms of the radius of the ball.



1) $\frac{2r}{5}$ **2)** $\frac{3r}{5}$ **3)** 0 **4)** $\frac{r}{2}$

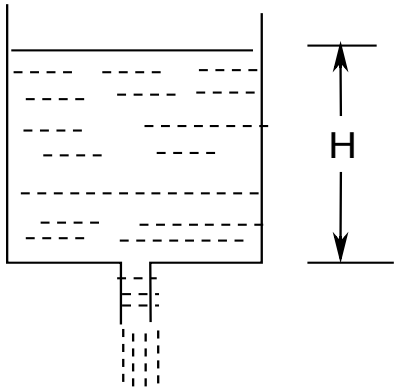
33. An ice cube is floating in water above which a layer of a lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively



1) rise and fall **2)** fall and rise

3) not change and not change **4)** not change and fall

34. A tank of cross sectional area A_0 is filled with a liquid. A small orifice of area A $\left(A << A_0\right)$ is present at the bottom of tank. At a moment the height of liquid in the tank is 'H', what is acceleration of top layer of liquid at this moment? (viscosity of liquid negligible)



1) g **2)** $\frac{A}{A_0}g$ **3)** $\left(\frac{A}{A_0}\right)^2 g$ **4)** depends on H

35. Density of a planet is two times the density of earth. Radius of this planet is half. Match, the followings. (As compared to earth) Assuming default reference for potential energy.

Column – I		**Column – II**	
A	Acceleration due to gravity on this planet's surface	p	Half
B	Gravitational potential on the surface	q	Same
C	Gravitational potential at centre	r	Two times
D	Gravitational field strength at centre	s	Four times

1) A – Q;B – P; C – P;D – Q **2)** A – Q; B–R; C – P;D–Q

3) A – Q;B – P;C – Q;D – P **4)** A – Q; B–R; C – P;D–S

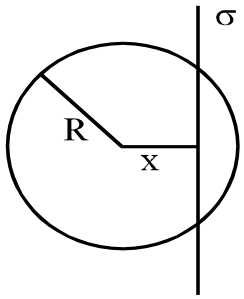
36. An ideal gas undergoes a process in which co-efficient of volume expansion of gas , varies with absolute temperature by the relation $\gamma = \frac{2}{T}$ Let C is the molar heat capacity in this process and C_P, C_V are molar heat capacity at constant pressure and volume respectively. Then

1) $C = 3C_V + 2R$ **2)** $C = 2C_V + R$ **3)** $C = C_V + 2R$ **4)** $C = C_V - R$

37. A hollow sphere of inner radius R and outer radius 2R is made from a material of thermal conductivity K. It is surrounded by another hollow sphere of inner radius 2R and outer radius 3R made by material of thermal conductivity 2K. The inside of smaller sphere and outside of bigger sphere are maintained at 0ºC and 70ºC respectively. Temperature of interface in steady state, will be

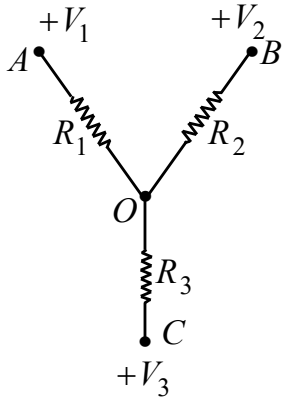
1) 20º C **2)** 40º C **3)** 50º C **4)** 60º C

38. An infinite, uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux through the Gaussian surface is



1) $\frac{\pi R^2 \sigma}{\varepsilon_0}$ **2)** $\frac{2\pi\left(R^2 - x^2\right)\sigma}{\varepsilon_0}$ **3)** $\frac{\pi(R-x)^2\sigma}{\varepsilon_0}$ **4)** $\frac{\pi\left(R^2 - x^2\right)\sigma}{\varepsilon_0}$

39. Three resistances are joined together to from a letter *Y,* as shown in fig. if the potentials of the terminals *A, B* and *C* are V_1, V_2 and V_3 respectively, then determine the potential of the point 'O'.



1) $\left[\frac{V_1}{R_1^2}+\frac{V_2}{R_2^2}+\frac{V_3}{R_3^2}\right]\left[\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right]^2$ **2)** $\left[\frac{V_1}{R_1}+\frac{V_2}{R_2}+\frac{V_3}{R_3}\right]\left[\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right]^{-1}$

3) $\left[\frac{V_1}{R_1}+\frac{V_2}{R_2}+\frac{V_3}{R_3}\right]\left[R_1+R_2+R_3\right]$ **4)** $\left[\frac{V_1}{R_1^2}+\frac{V_2}{R_2^2}+\frac{V_3}{R_3^2}\right]\left[R_1^2+R_2^2+R_3^2\right]$

40. **Assertion :**Potential difference across the battery is always equal to the emf of the battery.

Reason: Work done by the battery per unit charge is the emf of the battery

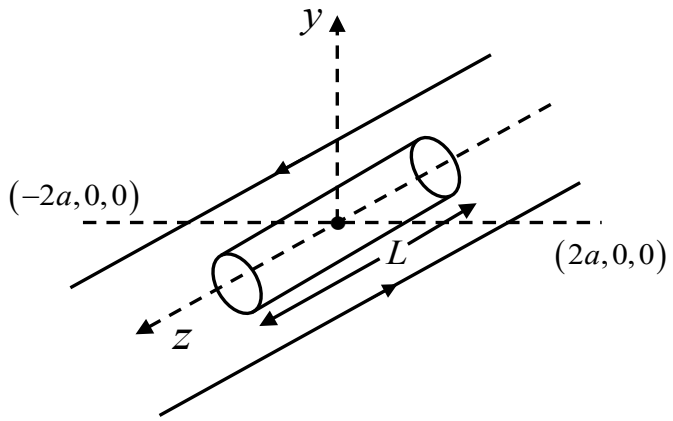
1) Assertion is True, Reason is True; Reason is **a Correct** explanation for Assertion

2) Assertion is True, Reason is True; Reason is **a not correct** explanation for Assertion

3) Assertion is True, Reason is False

4) Assertion is False, Reason is True

41. Two very long wires parallel to the z-axis (in xy plane)and a distance 4a (along x-axis) apart carry equal currents I in apposite directions as shown in the figure. A cylinder of radius a and length L has its axis on the z-axis midway between the wires. Calculate the net upward magnetic flux through half of the curved cylindrical surface above the xz plane

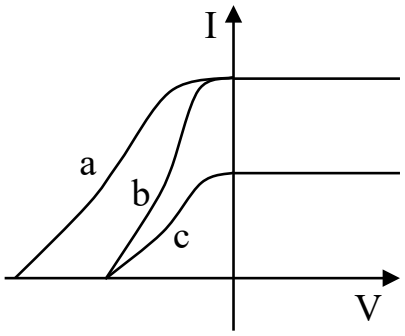


1) $\frac{\mu_0 IL}{\pi}$ **2)** $\frac{\mu_0 IL \ln 3}{\pi}$ **3)** $\frac{\mu_0 IL \ln 2}{\pi}$ **4)** $\frac{\mu_0 IL}{2}$

42. A disc of resistivity ρ, radius *a,* and thickness *b* is placed with its plane perpendicular to a time varying field, $B = B_0 \sin \omega t$ circular currents called eddy currents are induced, and energy is dissipated in the disc. The average power dissipation in the disc is

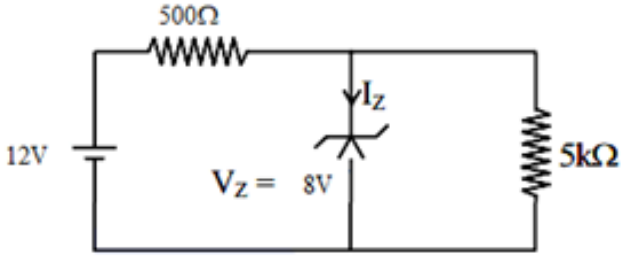
1) $\frac{1}{8\rho}\pi b a^2 \omega^2 B_0^2$ **2)** $\frac{1}{4\rho}\pi b a^2 \omega^2 B_0^2$ **3)** $\frac{1}{8\rho}\pi b a^4 \omega^2 B_0^2$ **4)** $\frac{1}{16\rho}\pi b a^4 \omega^2 B_0^2$

43. In a photo electric experiment I(current)-V (voltage) graph is as shown. Curves a,b,c correspond to three different metal surfaces irradiated with monochromatic light of same frequency. Assuming the ratio of number of electrons emitted per second to the number of photons incident per second is the same for all the three surfaces, choose the INCORRECT statement :



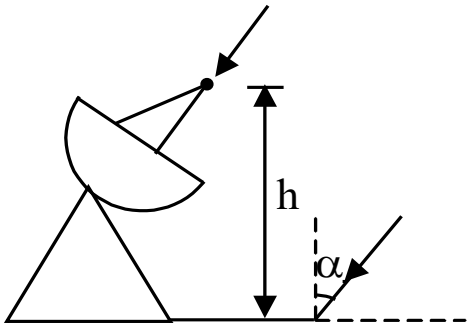
1) the work function of metals b and c are equal
2)the intensities of light incident on a and b are same
3)the work functions of metals a and b are not equal
4)the intensities of light incident on a, b and c are all different

44. The power lost in a zener diode in the given circuit is P mw then P =



1)51.2 **2)**53.2 **3)**55.2 **4)**49.2

45. Radio waves coming at angle α to vertical are received by a ladder after reflection from a nearby water surface directly. What can be height of antenna from water surface so that it records a maximum intensity (a maxima) (wavelength $=\lambda$).



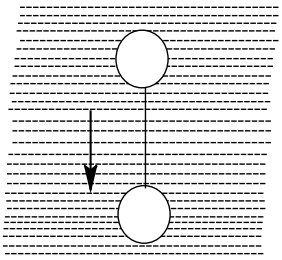
1) $\frac{\lambda}{2\cos\alpha}$ **2)** $\frac{\lambda}{2\sin\alpha}$ **3)** $\frac{\lambda}{4\sin\alpha}$ **4)** $\frac{\lambda}{4\cos\alpha}$

SECTION-II(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** andIf answer is from **10.5** and less than **11** round off is **11).**

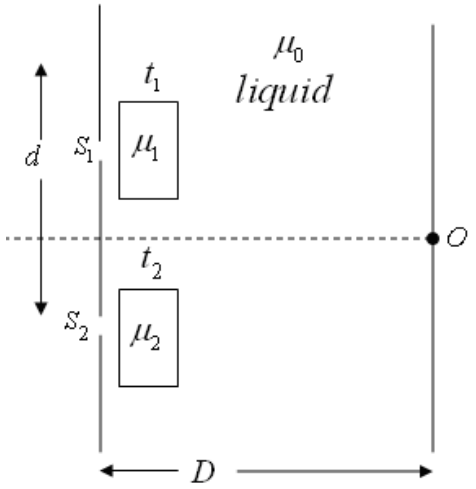
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. Figure shows a system of two balls of same volume $=0.25m^3$ moving with terminal velocity in a long uniform viscous medium (negligible density). Balls are connected with a thin metal string of cross-section area $0.5cm^2$. Density of upper and lower ball are $1gm/cm^3$ and $9gm/cm^3$ respectively. In this situation metallic string is just about to break. Ultimate stress of material of a string _____ $\times 10^8 N/m^2$. (g = $10m/s^2$. Coefficient of viscosity is same for both bodies, ignore atmospheric pressure.)

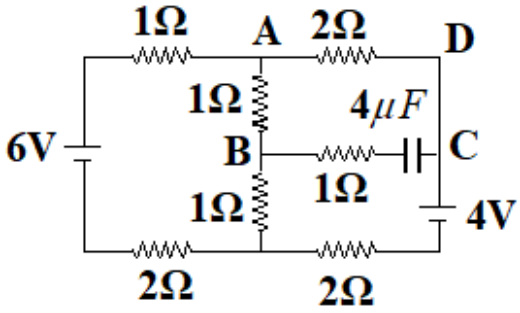


47. A uniform elastic plank moves due to a constant force $F_0 = 84N$ distributed uniformly over the end face of the plank. The area of the end face is $10cm^2$ and young's modulus is $2.1\times10^{11} N/m^2$. If the strain produced in the direction of the force is $x\times10^{-7}$, then find the value of x.

48. A liquid of refractive index μ_0 is filled between slits and screen. Initially the central maxima is at point O on screen, with no slabs in front of slits. Now two slabs of refractive index μ_1 and μ_2, with thickness t_1 and t_2 respectively are placed in front of slits as shown, So that the position of central maxima does not change. Find the ratio of $\frac{t_1}{t_2}$, if $\mu_0 = 1.5,\ \mu_1 = 1.8\ and\ \mu_2 = 2.1.$



49. Find the charge in μC on capacitor in steady state



50. A rod is moving with velocity 9 m/sec perpendicular to external magnetic field of 1T and there is a conducting sphere having charge 30C and radius R. If electric field inside rod is same as electric field on the surface of sphere and radius $R = x\times10^5$ m . Find the value of x (Round of to nearest integer)

CHEMISTRY **Max Marks: 100**

SECTION-I(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Alum used in the purification of drinking water in Nalgonda by using NEERI Technology then find equivalent weight of Alum

1) $\frac{M}{8}$ **2)** $\frac{M}{5}$ **3)** $\frac{M}{6}$ **4)** $\frac{M}{24}$

52. Find the total number of chiral carbon atoms in the products when the carbohydrate of the sucrose and maltose on hydrolysis is p then $\frac{p}{2}$ is

1) 10 **2) 8** **3) 12** **4) 6**

53. A sample of 0.5g of an organic compound was treated accounting to Kjeldahl's method. The ammonia evolved was absorbed in 50ml of 0.5M H_2SO_4. The residual acid required 60ml of 0.5M solution of caustic soda for neutralization. Find the percentage composition of nitrogen in the organic compound

1) 60 **2)** 72 **3)** 56 **4)** 86

54. Given below are two statements. One is labelled as assertion (A) and the other is labelled as reason (R)

Assertion: $T\ell I_3$ is isomorphous to CsI_3, the metal is present in +1 oxidation state

Reason: $T\ell$ metal has fourteen – f-electrons in its electronic configuration

1) Assertion is correct but reason not correct

2) Assertion is not correct but reason is correct

3) Both assertion and reason are correct and reason is correct explanation of assertion

4) Both assertion and reason are correct but reason is not correct explanation of assertion

55. 5g sample contain only Na_2CO_3 and Na_2SO_4. This sample is dissolved and the volume made up to 250ml. 25ml of this solution neutralizes 20ml of 0.1M H_2SO_4. Calculate the percentage of Na_2SO_4 in the given sample

1) 57.6 **2)** 50.6 **3)** 53.6 **4)** 42.6

56. **Statement1:** Except Boron all other elements of 13th group are soft metals

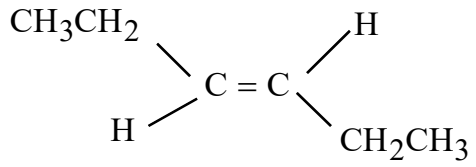
Statement 2: Electronegativity of B > Al < Ga < In < Tl

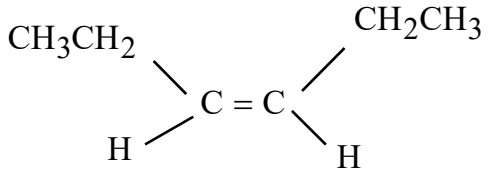
1) S1 and S2 are true **2)** S1 false S2 true

3) S1 and S2 are false **4)** S1 true S2 false

57. $\text{Butyne} \xrightarrow{NaNH_2} A \xrightarrow{CH_3CH_2Br} B \xrightarrow[Li, NH_{3(\ell)}]{H_2} C$ C in the above reaction is

1) $CH_3CH_2CH=CH-CH_3$ **2)** $CH_3CHCH=CH_2$

3) 

4) 

58. Identify the incorrect match

Name	**IUPAC official name**
a) Unnilunium	i) mendalevium
b) Unniltrium	ii) Law rencium
c) Unnilhexium	iii) seaborgium
d) Un ununnium	iv) Darmastadtium

1) b, ii **2)** c, iii **3)** d, iv **4)** a, i

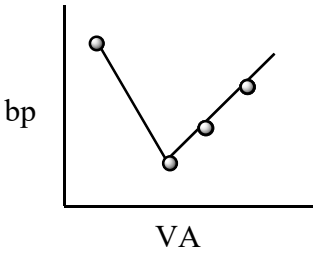
59. **Statement 1:** 0.1 mole of permanganate ion in acidic medium can oxididise 0.5 mole of Fe^{2+}

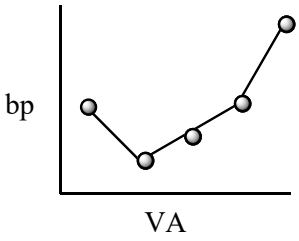
Statement 2: 0.1 mole of permanganate ion in acidic medium oxidise 0.166 mol of FeC_2O_4

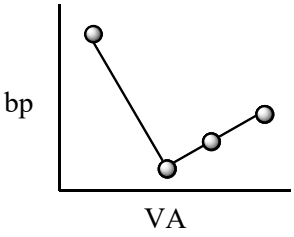
1) S1 true S2 false **2)** S1 and S2 are correct

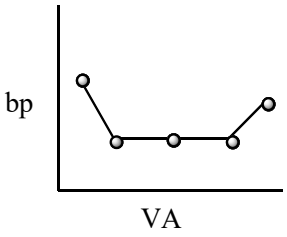
3) S1 false S2 true **4)** S1 and S2 are false

60. Which of the following Graphs indicates correct about VA group boiling point of hydrides

1) 

2) 

3) 

4) 

61. **Assertion:** p – nitrophenol is more acidic than ortho and meta nitrophenol

Reason: Methyl carbinol will give immediate turbidity with Lucas reagent

1) A correct R is not correct **2)** A is correct R is correct

3) A not correct R is correct **4)** A not correct R is not correct

62. 1.2 ml of Acetic acid having density 1.06g/mL is dissolved in 2000 ml of water. The depression in freezing point was 0.0205°C then find van'thoff factor and proton concentration of acid solution

1) $1.041, 4.24\times10^{-4}\,m/L$ **2)** $1.1041, 5.3\times10^{-4}\,m/L$

3) $1.099, 6.2\times10^{-4}\,m/L$ **4)** $10.411, 3.23\times10^{-4}\,m/L$

63. The experimental data for the reaction $2A + B_2 \rightarrow 2AB$ is given below, based on the given data find rate equation

[A] mol /L	**[B_2] mol /L**	**rate mol /L – sec**
0.5	0.5	1.6×10^{-4}
0.5	1.0	3.2×10^{-4}
1.0	1.0	3.2×10^{-4}

1) $\text{rate} = K[A]^0[B_2]^1$ **2)** $\text{rate} = K[A]^1[B_2]^0$

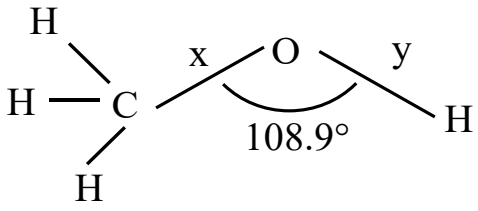
3) $\text{rate} = K[A][B_2]$ **4)** $\text{rate} = K[A]^{1/2}[B_2]^1$

64. The quantity of electricity needed to reduce one centimole of dichromate ion in acidic medium to chromic state

1) 579 coulombs **2)** 5790 coulombs

3) 57900 coulombs **4)** 579000 coulombs

65. The sum of the x, y bond length is



1) $x = 142pm, y = 96pm\; x + y = 248pm$ **2)** $x = 96pm, y = 142pm\; x + y = 298pm$

3) $x = 165pm, y = 69pm\; x + y = 234pm$ **4)** $x = 122pm, y = 80pm\; x + y = 202pm$

66. Vitamins are Fat soluble and water soluble, maximum all the water soluble vitamins are excreted in urine and cannot be stored, but the vitamin which cannot be exerted inurine is

1) A - Retinol **2)** B_6 - pyridoxine

3) cyanocobalamine B_{12} **4)** Folic acid B_9 and Ascorbicacid (C)

67. Match the following

	Amino acid		**One letter code**
i.	Tryptophan	P.	A
ii.	Lysine	Q.	W
iii.	Arginine	R.	T
iv.	Phenyl alanine	S.	K
		T.	F
		U.	R

1) i-R; ii-U; iii-P; iv-Q **2)** i-T; ii-Q; iii-P; iv-R

3) i-Q; ii-S; iii-U; iv-T **4)** i-Q; ii-T; iii-P; iv-S

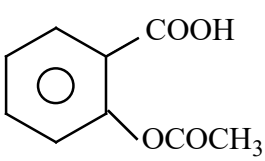
68. In the following sequence of reaction Toluene $\xrightarrow{KMnO_4}$ A $\xrightarrow[\text{chloride}]{\text{Thionyl}}$ B $\xrightarrow[BaSO_4]{H_2/Pd}$ C

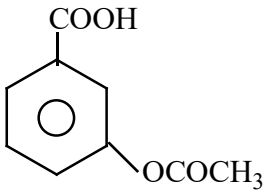
Product C is

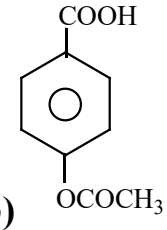
1) Benzoic acid **2)** Benzyl alcohol

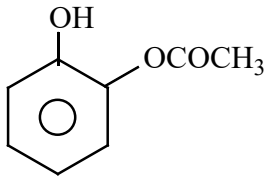
3) Methyl benzene **4)** Benzene carbaldehyde

69. Sodium phenoxide when heated with CO_2 under pressure at 125°C yields a product which on acylation produces C. Major product C is

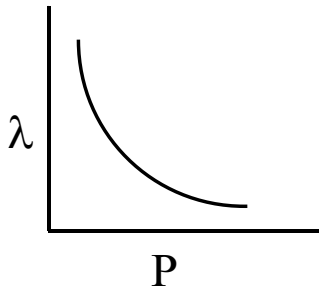
1) 

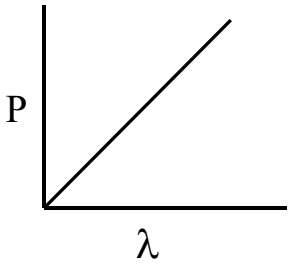
2) 

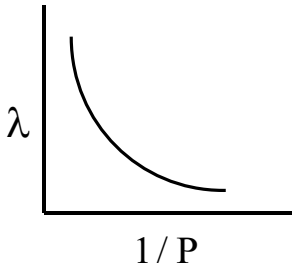
3) 

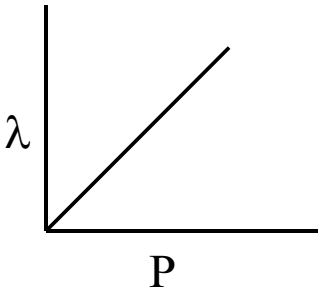
4) 

70. According to the wave particle duality of matter by de-Broglie, which of the following graph plot pressure most appropriate relationship between wavelength of electron and momentum of electron (P)?

1) 

2) 

3) 

4) 

SECTION-II(NUMERICAL VALUE TYPE)

This section contains **5Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** andIf answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Total number of carbon atoms present in cysteine an amino acid

72. The number of compounds is/are not coloured yellow

(i) $Zn_2[Fe(CN)_6]$ (ii) $K_3[Co(NO_2)_6]$ (iii) $(NH_4)_3[As(MO_3O_{10})_4]$ (iv) $BaCrO_4$,

(v) $Fe_4[Fe(CN)_6]_3$ (vi) $(NH_4)_3PO_4\ 12MoO_3$, (vii) CdS (viii) $PbCrO_4$ (ix) PbI_2.

73. The compound formed by the reaction of Propanal with semi carbazide contains number of nitrogen atoms is x and number of oxygen is y then x + y is

74. The total number of sigma and Pi bonds in 2 – oxohex-4-ynoic acid is x and no. of hybrid orbitals in the compound hex-1,3,5-triene is y. Then x + y is

75. Carbolic acid on concentrated H_2SO_4 forms product A. Which on concentrated HNO_3 gives product B. Then find sum of oxygen atoms in A and B are……

Sri Chaitanya
Educational Institutions & Techno Schools

Infinity Learn

39 Years of Excellence 1986-2024



1 ALL INDIA RANK

K C BASAVA REDDY
APPL.No. 240310618179*

EMPOWERING EVERY STUDENT
TO BECOME EXTRAORDINARY

PROUDLY ACHIEVED 222 RANKS IN TOP 1000

SEIZES 4 RANKS IN TOP 10 IN ALL-INDIA RANKS



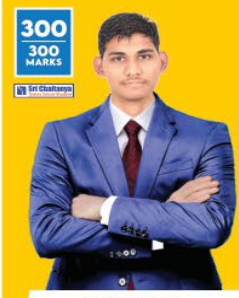
ALL INDIA RANK **3**

THOTAMSETTY NIKILESH
APPL.No. 240310813888* — RANK



ALL INDIA RANK **6**

HIMANSHU THALOR
APPL.No. 240310580429* — RANK



ALL INDIA RANK **9**

REDDI ANIL
APPL.No.240310238514 — RANK

SECURED 25 RANKS IN TOP 100
ALL INDIA OPEN CATEGORY

Sri Chaitanya - Nagpur
DLP Student

1 RANK	9 RANK	14 RANK	20 RANK	21 RANK	22 RANK	26 RANK	28 RANK		
G N NIRMALKUMAR Appl.No. 240310385062*	REDDI ANIL Appl.No. 240310238514*	K C BASAVA REDDY Appl.No. 240310618179*	THOTAMSETTY NIKILESH Appl.No. 240310813888*	A V TANISH REDDY Appl.No. 240310807613	HIMANSHU THALOR Appl.No. 240310580429**	VEDANT SAINI Appl.No. 240310182830	P MEET VIKRAMBHAI Appl.No. 240310157524*		
34 RANK	40 RANK	43 RANK	46 RANK	49 RANK	52 RANK	53 RANK	57 RANK		
SANVI JAIN Appl.No. 240310150036*	VISHARAD SRIVASTAVA Appl.No. 240310046262*	T JAYADEV REDDY Appl.No. 240310167365	ISHAAN GUPTA Appl.No. 240310100229*	MAVURU JASWITH Appl.No. 240310542275*	DORISALA SRINIVASA REDDY Appl.No. 240310682440*	ARCHIT RAHUL PATIL Appl.No. 240310512311*	KRISHNA AGRAWAL Appl.No. 240310285850*		
60 RANK	68 RANK	70 RANK	76 RANK	92 RANK	93 RANK	95 RANK	96 RANK	98 RANK	
AYUSH GANGAL Appl.No. 240310270709	PALAGIRI SATHISH REDDY Appl.No. 240310905497	MD K GHOUSE MOHIUDDIN Appl.No. 240310176352	T V S SAI NAGA BHUSHAN Appl.No. 240310868568	M M PRUTHVI RAJ Appl.No. 240311084545	M SAI SIVA LOCHAN Appl.No. 240310866829*	RAJDEEP MISHRA Appl.No. 240310265621*	MANOJ SOHAN GAJULA Appl.No. 240310529661	KRISH NARSARIA Appl.No. 240310028286*	



Below 100 All-India Open Category Ranks — 25

Below 500 All-India Open Category Ranks — 108

Below 1000 All-India Open Category Ranks — 222

Below 100 All-India All Category Ranks — 97

Below 1000 All Category Ranks — 888

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2024 21,987

FOR OFFER ON JEE MAIN & JEE ADVANCED COURSES

SCAN THE QR CODE

www.srichaitanya.net | Ph: 040 660 60606

* DLP | @ Gen-EWS | $ OBC-(NCL) | % SC | # ST | & GEN